

**Angelo Cantoli - Simone Edmondo
Bedini**



POLITECNICO
MILANO 1863

Requirement Analysis and Specification Document

Deliverable:	RASD
Title:	Requirement Analysis and Verification Document
Authors:	Angelo Cantoli, Simone Edmondo Bedini
Version:	1.0
Date:	28/10/2025
Download page:	https://github.com/angelo-cantoli/CantoliBedini
Copyright:	Copyright © 2025 - Angelo Cantoli, Simone Edmondo Bedini - All rights reserved

Contents

Table of Contents	3
List of Figures	4
List of Tables	4
1 Introduction	5
1.1 Purpose	5
1.1.1 Goals	5
1.2 Scope	5
1.2.1 World Phenomena	5
1.2.2 Shared Phenomena	6
1.3 Definitions, Acronyms, Abbreviations	6
1.4 Revision History	6
1.5 Reference Documents	6
1.6 Document Structure	6
2 Overall Description	7
3 Specific Requirements	8
4 Formal Analysis Using Alloy	9
5 Effort Spent	10
References	11

List of Figures

List of Tables

1 Introduction

1.1 Purpose

BBP is a software designed to simplify the life of everyone who goes on bike trips and wants to enjoy them without worrying about finding the best routes.

It is intended as a community for cyclists, in which every person can share information about their personal routes, so that everyone may take advantage of the information and experience of other bikers. The application is intended for smartphone, where it can make use of GPS and sensor data to seamlessly reconstruct the path a cyclist has taken.

1.1.1 Goals

- G1: Cyclists maintain accurate and comprehensive records of their bike trips
 - G1.1: Bike trip data is automatically tracked while cycling
 - G1.2: Recording bike trips remains possible without automatic tracking
 - G1.3: The records of cyclists' bike trips are persistent and always accessible
 - G1.4: Cyclists' performance statistics are observable throughout their trip history
 - G1.5: Recorded bike trips include the condition of the roads on the bike path
 - G1.6: Hazards and obstacles on the bike path are identified and recorded
 - G1.7: Recorded bike trips include meteorological conditions when available
- G2: Cyclists make informed route selections
 - G2.1: All known paths that cover a route are observable
 - G2.2: The most suitable paths between two locations are distinguishable
 - G2.3: Path suitability is objectively evaluable based on road condition and route effectiveness
- G3: A community knowledge base of bike path information is maintained by cyclists
 - G3.1: Cyclists contribute bike path information by sharing their recorded trips
 - G3.2: Published bike path information is accessible to all users regardless of registration

1.2 Scope

1.2.1 World Phenomena

- WP1: User goes cycling along roads
- WP2: Roads contain potholes or other obstacles
- WP3: Road conditions vary over time
- WP4: Meteorological conditions change over time
- WP5: User needs to find routes between two locations
- WP6: User decides to record a bike trip
- WP7: Cyclists decide to share information with a community of other cyclists

1.2.2 Shared Phenomena

- SP1: User creates an account
- SP2: User logs into their account
- SP3: User logs out of their account
- SP4: User starts trip recording
- SP5: User stops trip recording
- SP6: System confirms trip has been saved
- SP7: System displays statistics regarding saved trips
- SP8: System retrieves meteorological information from an external service
- SP9: System displays trip information enriched with weather data
- SP10: User manually inserts street names and status of a bike path
- SP11: User manually inserts the presence of an obstacle on the path
- SP12: User enables the system to automatically acquire data from their mobile device
- SP13: System detects if the user is biking based on speed data
- SP14: Mobile device's GPS provides location coordinates for path reconstruction
- SP15: Mobile device's accelerometer provides motion data
- SP16: Mobile device's gyroscope provides orientation data
- SP17: User confirms automatically detected obstacles
- SP18: User corrects automatically detected obstacles
- SP19: User publishes bike trip information
- SP20: User specifies an origin and destination for path visualization
- SP21: System displays relevant path information
- SP22: System displays paths between two locations, ranked based on score
- SP23: User views their recorded trips
- SP24: User edits one of their recorded trips
- SP25: User deletes one of their recorded trips

1.3 Definitions, Acronyms, Abbreviations

1.4 Revision History

1.5 Reference Documents

1.6 Document Structure

2 Overall Description

3 Specific Requirements

4 Formal Analysis Using Alloy

5 Effort Spent

References